INCH-POUND

MIL-DTL-25988/1B
29 June 2021

SUPERSEDING
MIL-R-25988/1A
10 July 1975

DETAIL SPECIFICATION SHEET

RUBBER, FLUOROSILICONE ELASTOMER, OIL- AND FUEL-RESISTANT, O-RINGS, CLASS 1, GRADE 70

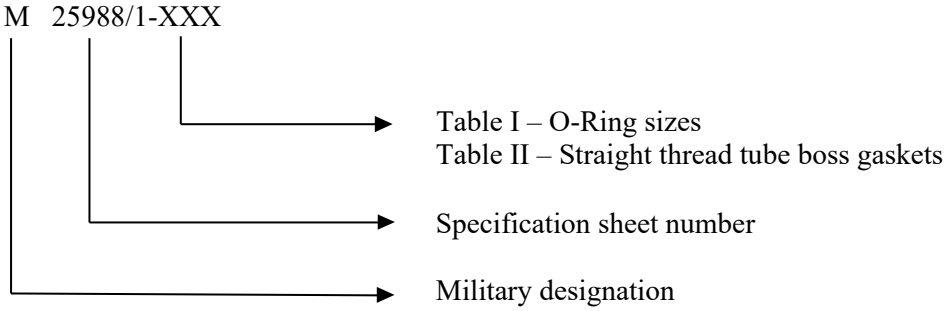
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-25988.

REQUIREMENTS:

Scope. This specification sheet describes a standardized part numbering system for the procurement of fluorosilicone elastomer rubber oil and fuel resistant o-rings Class 1, Grade 70, intended for use in packing glands as specified in MIL-G-5514.

Part or identifying number (PIN). The PIN shall consist of the prefix M25988/1, followed by the assigned dash number for o-rings and boss gaskets.



AMSC N/A FSC 5331

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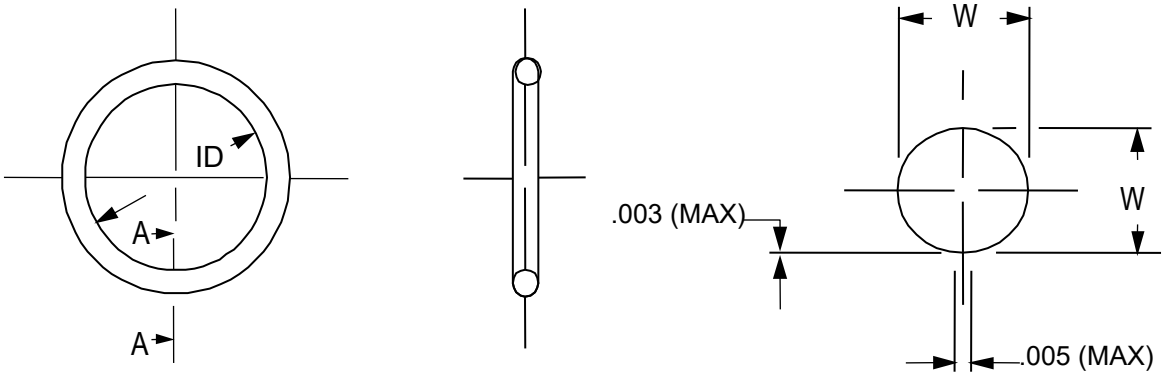


FIGURE 1. Dimensions of shapes cut from sheets and strips.

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TABLE I. O-Ring sizes.

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
001	.040	±.003	.029	±.004
002	.050	±.003	.042	±.004
003	.060	±.003	.056	±.004
004	.070	±.003	.070	±.005
005	.070	±.003	.101	±.005
006	.070	±.003	.114	±.005
007	.070	±.003	.145	±.005
008	.070	±.003	.176	±.005
009	.070	±.003	.208	±.005
010	.070	±.003	.239	±.005
011	.070	±.003	.301	±.005
012	.070	±.003	.364	±.005
013	.070	±.003	.426	±.005
014	.070	±.003	.489	±.005
015	.070	±.003	.551	±.007
016	.070	±.003	.614	±.009
017	.070	±.003	.676	±.009
018	.070	±.003	.739	±.009
019	.070	±.003	.801	±.009
020	.070	±.003	.864	±.009
021	.070	±.003	.926	±.009
022	.070	±.003	.989	±.010
023	.070	±.003	1.051	±.010
024	.070	±.003	1.114	±.010
025	.070	±.003	1.176	±.011
026	.070	±.003	1.239	±.011
027	.070	±.003	1.301	±.011
028	.070	±.003	1.364	±.013
029	.070	±.003	1.489	±.013
030	.070	±.003	1.614	±.013
031	.070	±.003	1.739	±.015
032	.070	±.003	1.864	±.015
033	.070	±.003	1.989	±.018
034	.070	±.003	2.114	±.018
035	.070	±.003	2.239	±.018
036	.070	±.003	2.364	±.018

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
037	.070	±.003	2.489	±.018
038	.070	±.003	2.614	±.020
039	.070	±.003	2.739	±.020
040	.070	±.003	2.864	±.020
041	.070	±.003	2.989	±.024
042	.070	±.003	3.239	±.024
043	.070	±.003	3.489	±.024
044	.070	±.003	3.739	±.027
045	.070	±.003	3.989	±.027
046	.070	±.003	4.239	±.030
047	.070	±.003	4.489	±.030
048	.070	±.003	4.739	±.030
049	.070	±.003	4.989	±.037
050	.070	±.003	5.239	±.037
102	.103	±.003	.049	±.005
103	.103	±.003	.081	±.005
104	.103	±.003	.112	±.005
105	.103	±.003	.143	±.005
106	.103	±.003	.174	±.005
107	.103	±.003	.206	±.005
108	.103	±.003	.237	±.005
109	.103	±.003	.299	±.005
110	.103	±.003	.362	±.005
111	.103	±.003	.424	±.005
112	.103	±.003	.487	±.005
113	.103	±.003	.549	±.007
114	.103	±.003	.612	±.009
115	.103	±.003	.674	±.009
116	.103	±.003	.737	±.009
117	.103	±.003	.799	±.010
118	.103	±.003	.862	±.010
119	.103	±.003	.924	±.010
120	.103	±.003	.987	±.010
121	.103	±.003	1.049	±.010
122	.103	±.003	1.112	±.010

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TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
123	.103	±.003	1.174	±.012
124	.103	±.003	1.237	±.012
125	.103	±.003	1.299	±.012
126	.103	±.003	1.362	±.012
127	.103	±.003	1.424	±.012
128	.103	±.003	1.487	±.012
129	.103	±.003	1.549	±.015
130	.103	±.003	1.612	±.015
131	.103	±.003	1.674	±.015
132	.103	±.003	1.737	±.015
133	.103	±.003	1.799	±.015
134	.103	±.003	1.862	±.015
135	.103	±.003	1.925	±.017
136	.103	±.003	1.987	±.017
137	.103	±.003	2.050	±.017
138	.103	±.003	2.112	±.017
139	.103	±.003	2.175	±.017
140	.103	±.003	2.237	±.017
141	.103	±.003	2.300	±.020
142	.103	±.003	2.362	±.020
143	.103	±.003	2.425	±.020
144	.103	±.003	2.487	±.020
145	.103	±.003	2.550	±.020
146	.103	±.003	2.612	±.020
147	.103	±.003	2.675	±.022
148	.103	±.003	2.737	±.022
149	.103	±.003	2.800	±.022
150	.103	±.003	2.862	±.022
151	.103	±.003	2.987	±.024
152	.103	±.003	3.237	±.024
153	.103	±.003	3.487	±.024
154	.103	±.003	3.737	±.028
155	.103	±.003	3.987	±.028
156	.103	±.003	4.237	±.030
157	.103	±.003	4.487	±.030
158	.103	±.003	4.737	±.030

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
159	.103	±.003	4.987	±.035
160	.103	±.003	5.237	±.035
161	.103	±.003	5.487	±.035
162	.103	±.003	5.737	±.035
163	.103	±.003	5.987	±.035
164	.103	±.003	6.237	±.035
165	.103	±.003	6.487	±.040
166	.103	±.003	6.737	±.040
167	.103	±.003	6.987	±.040
168	.103	±.003	7.237	±.045
169	.103	±.003	7.487	±.045
170	.103	±.003	7.737	±.045
171	.103	±.003	7.987	±.045
172	.103	±.003	8.237	±.050
173	.103	±.003	8.487	±.050
174	.103	±.003	8.737	±.050
175	.103	±.003	8.987	±.050
176	.103	±.003	9.237	±.055
177	.103	±.003	9.487	±.055
178	.103	±.003	9.737	±.055
201	.139	±.004	0.171	±.005
202	.139	±.004	0.234	±.005
203	.139	±.004	0.296	±.005
204	.139	±.004	0.359	±.005
205	.139	±.004	0.421	±.005
206	.139	±.004	0.484	±.005
207	.139	±.004	0.546	±.007
208	.139	±.004	0.609	±.009
209	.139	±.004	0.671	±.009
210	.139	±.004	0.734	±.010
211	.139	±.004	0.796	±.010
212	.139	±.004	0.859	±.010
213	.139	±.004	0.921	±.010
214	.139	±.004	0.984	±.010
215	.139	±.004	1.046	±.010

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TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
216	0.139	±.004	1.109	±.012
217	0.139	±.004	1.171	±.012
218	0.139	±.004	1.234	±.012
219	0.139	±.004	1.296	±.012
220	0.139	±.004	1.359	±.012
221	0.139	±.004	1.421	±.012
222	0.139	±.004	1.484	±.015
223	0.139	±.004	1.609	±.015
224	0.139	±.004	1.734	±.015
225	0.139	±.004	1.859	±.018
226	0.139	±.004	1.984	±.018
227	0.139	±.004	2.109	±.018
228	0.139	±.004	2.234	±.020
229	0.139	±.004	2.359	±.020
230	0.139	±.004	2.484	±.020
231	0.139	±.004	2.609	±.020
232	0.139	±.004	2.734	±.024
233	0.139	±.004	2.859	±.024
234	0.139	±.004	2.984	±.024
235	0.139	±.004	3.109	±.024
236	0.139	±.004	3.234	±.024
237	0.139	±.004	3.359	±.024
238	0.139	±.004	3.484	±.024
239	0.139	±.004	3.609	±.028
240	0.139	±.004	3.734	±.028
241	0.139	±.004	3.859	±.028
242	0.139	±.004	3.984	±.028
243	0.139	±.004	4.109	±.028
244	0.139	±.004	4.234	±.030
245	0.139	±.004	4.359	±.030
246	0.139	±.004	4.484	±.030
247	0.139	±.004	4.609	±.030
248	0.139	±.004	4.734	±.030
249	0.139	±.004	4.859	±.035
250	0.139	±.004	4.984	±.035
251	0.139	±.004	5.109	±.035

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
252	0.139	±.004	5.234	±.035
253	0.139	±.004	5.359	±.035
254	0.139	±.004	5.484	±.035
255	0.139	±.004	5.609	±.035
256	0.139	±.004	5.734	±.035
257	0.139	±.004	5.859	±.035
258	0.139	±.004	5.984	±.035
259	0.139	±.004	6.234	±.040
260	0.139	±.004	6.484	±.040
261	0.139	±.004	6.734	±.040
262	0.139	±.004	6.984	±.040
263	0.139	±.004	7.234	±.045
264	0.139	±.004	7.484	±.045
265	0.139	±.004	7.734	±.045
266	0.139	±.004	7.984	±.045
267	0.139	±.004	8.234	±.050
268	0.139	±.004	8.484	±.050
269	0.139	±.004	8.734	±.050
270	0.139	±.004	8.984	±.050
271	0.139	±.004	9.234	±.055
272	0.139	±.004	9.484	±.055
273	0.139	±.004	9.734	±.055
274	0.139	±.004	9.984	±.055
275	0.139	±.004	10.484	±.055
276	0.139	±.004	10.984	±.065
277	0.139	±.004	11.484	±.065
278	0.139	±.004	11.984	±.065
279	0.139	±.004	12.984	±.065
280	0.139	±.004	13.984	±.065
281	0.139	±.004	14.984	±.065
282	0.139	±.004	15.955	±.075
283	0.139	±.004	16.955	±.080
284	0.139	±.004	17.955	±.085
309	0.21	±.005	0.412	±.005
310	0.21	±.005	0.475	±.005

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TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)				DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)			Width (W)		Inside Diameter (ID)	
311	.210	±.005	.537	±.007	347	.210	±.005	4.225	±.030
312	.210	±.005	.600	±.009	348	.210	±.005	4.350	±.030
313	.210	±.005	.662	±.009	349	.210	±.005	4.475	±.030
314	.210	±.005	.725	±.010	350	.210	±.005	4.600	±.030
315	.210	±.005	.787	±.010	351	.210	±.005	4.725	±.030
316	.210	±.005	.850	±.010	352	.210	±.005	4.850	±.030
317	.210	±.005	.912	±.010	353	.210	±.005	4.975	±.037
318	.210	±.005	.975	±.010	354	.210	±.005	5.100	±.037
319	.210	±.005	1.037	±.010	355	.210	±.005	5.225	±.037
320	.210	±.005	1.100	±.012	356	.210	±.005	5.350	±.037
321	.210	±.005	1.162	±.012	357	.210	±.005	5.475	±.037
322	.210	±.005	1.225	±.012	358	.210	±.005	5.600	±.037
323	.210	±.005	1.287	±.012	359	.210	±.005	5.725	±.037
324	.210	±.005	1.350	±.012	360	.210	±.005	5.850	±.037
325	.210	±.005	1.475	±.015	361	.210	±.005	5.975	±.037
326	.210	±.005	1.600	±.015	362	.210	±.005	6.225	±.040
327	.210	±.005	1.725	±.015	363	.210	±.005	6.475	±.040
328	.210	±.005	1.850	±.015	364	.210	±.005	6.725	±.040
329	.210	±.005	1.975	±.018	365	.210	±.005	6.975	±.040
330	.210	±.005	2.100	±.018	366	.210	±.005	7.225	±.045
331	.210	±.005	2.225	±.018	367	.210	±.005	7.475	±.045
332	.210	±.005	2.350	±.018	368	.210	±.005	7.725	±.045
333	.210	±.005	2.475	±.020	369	.210	±.005	7.975	±.045
334	.210	±.005	2.600	±.020	370	.210	±.005	8.225	±.050
335	.210	±.005	2.725	±.020	371	.210	±.005	8.475	±.050
336	.210	±.005	2.850	±.020	372	.210	±.005	8.725	±.050
337	.210	±.005	2.975	±.024	373	.210	±.005	8.975	±.050
338	.210	±.005	3.100	±.024	374	.210	±.005	9.225	±.055
339	.210	±.005	3.225	±.024	375	.210	±.005	9.475	±.055
340	.210	±.005	3.350	±.024	376	.210	±.005	9.725	±.055
341	.210	±.005	3.475	±.024	377	.210	±.005	9.975	±.055
342	.210	±.005	3.600	±.028	378	.210	±.005	10.475	±.060
343	.210	±.005	3.725	±.028	379	.210	±.005	10.975	±.060
344	.210	±.005	3.850	±.028	380	.210	±.005	11.475	±.065
345	.210	±.005	3.975	±.028	381	.210	±.005	11.975	±.065
346	.210	±.005	4.100	±.028	382	.210	±.005	12.975	±.065

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TABLE I. O-Ring sizes – Continued.

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
383	.210	±.005	13.975	±.070
384	.210	±.005	14.975	±.070
385	.210	±.005	15.955	±.075
386	.210	±.005	16.955	±.080
387	.210	±.005	17.955	±.085
388	.210	±.005	18.955	±.090
389	.210	±.005	19.955	±.095
390	.210	±.005	20.955	±.095
391	.210	±.005	21.955	±.100
392	.210	±.005	22.940	±.105
393	.210	±.005	23.940	±.110
394	.210	±.005	24.940	±.115
395	.210	±.005	25.940	±.120
425	.275	±.006	4.475	±.033
426	.275	±.006	4.600	±.033
427	.275	±.006	4.725	±.033
428	.275	±.006	4.850	±.033
429	.275	±.006	4.975	±.037
430	.275	±.006	5.100	±.037
431	.275	±.006	5.225	±.037
432	.275	±.006	5.350	±.037
433	.275	±.006	5.475	±.037
434	.275	±.006	5.600	±.037
435	.275	±.006	5.725	±.037
436	.275	±.006	5.850	±.037
437	.275	±.006	5.975	±.037
438	.275	±.006	6.225	±.040
439	.275	±.006	6.475	±.040
440	.275	±.006	6.725	±.040
441	.275	±.006	6.975	±.040
442	.275	±.006	7.225	±.045
443	.275	±.006	7.475	±.045
444	.275	±.006	7.725	±.045
445	.275	±.006	7.975	±.045

DASH NO.	ACTUAL SIZE (A)			
	Width (W)		Inside Diameter (ID)	
446	.275	±.006	8.475	±.055
447	.275	±.006	8.975	±.055
448	.275	±.006	9.475	±.055
449	.275	±.006	9.975	±.055
450	.275	±.006	10.475	±.060
451	.275	±.006	10.975	±.060
452	.275	±.006	11.475	±.060
453	.275	±.006	11.975	±.060
454	.275	±.006	12.475	±.060
455	.275	±.006	12.975	±.060
456	.275	±.006	13.475	±.070
457	.275	±.006	13.975	±.070
458	.275	±.006	14.475	±.070
459	.275	±.006	14.975	±.070
460	.275	±.006	15.475	±.070
461	.275	±.006	15.955	±.075
462	.275	±.006	16.455	±.075
463	.275	±.006	16.955	±.080
464	.275	±.006	17.455	±.085
465	.275	±.006	17.955	±.085
466	.275	±.006	18.455	±.085
467	.275	±.006	18.955	±.090
468	.275	±.006	19.455	±.090
469	.275	±.006	19.955	±.095
470	.275	±.006	20.955	±.095
471	.275	±.006	21.955	±.100
472	.275	±.006	22.940	±.105
473	.275	±.006	23.940	±.110
474	.275	±.006	24.940	±.115
475	.275	±.006	25.940	±.120

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TABLE II. Straight thread tube fitting boss gaskets.

DASH NO.	TUBE OUTSIDE DIAMETER	WIDTH (W) CROSS-SECTION		Inside Diameter (ID)	
901	.094	0.056	±.003	.185	±.005
902	.125	0.064	±.003	.239	±.005
903	.188	0.064	±.003	.301	±.005
904	.250	0.072	±.003	.351	±.005
905	.312	0.072	±.003	.414	±.005
906	.375	0.078	±.003	.468	±.005
907	.438	0.082	±.003	.530	±.007
908	.500	0.087	±.003	.644	±.009
909	.562	0.097	±.003	.706	±.009
910	.625	0.097	±.003	.755	±.009
911	.688	0.116	±.004	.863	±.009
912	.750	0.116	±.004	.924	±.009
913	.812	0.116	±.004	.986	±.010
914	.875	0.116	±.004	1.047	±.010
916	1.000	0.116	±.004	1.171	±.010
918	1.125	0.116	±.004	1.355	±.012
920	1.250	0.118	±.004	1.475	±.014
924	1.500	0.118	±.004	1.720	±.014
928	1.750	0.118	±.004	2.090	±.018
932	2.000	0.118	±.004	2.337	±.018

Referenced documents. In addition to MIL-DTL-25988, this document references MIL-G-5514, Gland Design, Packings, Hydraulic, General Requirements for.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - OS
Air Force – 20
DLA – CC

Preparing activity:
Air Force – 20

Reviewers:
Army – MR
Navy – AS
Air Force - 71

Agent:
Air Force – 110
(Project 5331-2021-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:41Z
Check the source to verify that this is the current version before use.